

Internships and master's theses

Scope

- This pages lists internships and master thesis projects
- please create sub pages for each internship project and master thesis
- please extend the list of open projects, so we have a discussion base for for incoming requests

Available projects

- see also
 - [TRENTOS University Course \(@ TUM\) - WS2020 - Team Projects](#)
 - [Proposed FV projects](#)
- TRENTOS-M SDK Development Tools
 - Extend the CAmkES Visualization tool
 - see also  [SEOS-934 - Authenticate](#) to see issue details
 - prepared ground for a interactive CAmkES GUI Editor, could be Eclipse based
 - implement a Debugger and integrate this into a development environment
 - seL4 thread awareness
 - CAmkES awareness
 - apply the Genode DDE concept to a seL4/CAmkES system
 - implement a simulation environment running natively on Linux for
 - Continue the work from [Jiawei Gao - seL4 API emulation in Linux](#)
 - CAmkES (there was a student project years ago at Data61 for this, might be able to get this to work again)
 - TRENTOS-M (run everything natively, or bridge things into a QEMU or a board where a TRENTOS-M core runs)
 - allow writing application in C++ instead of C
 - Prototyped by [Unknown User \(thowit01\)](#)
 - allow writing application in better languages than C to allow secure programming on TRENTOS-M
 - Ada/Spark, see [AdaCore](#)
 - D (<https://dlang.org>)
 - Go (<https://golang.org>)
 - Rust (<https://www.rust-lang.org>)
 - Zig (<https://ziglang.org>). Seem there is some activity at UNSW, see https://github.com/ShoshinX/zig_sel4_wrapper
 - Lua, Python, Ruby or JavaScript interpreters running in a CAmkES component
 - Java Virtual Machine prototype as a TRENTOS-M component
 - Make build_env including the dependencies for visual camkes  [SEOS-1961 - Authenticate](#) to see issue details
- TRENTOS-M Modules and Applications
 - Port <https://bitbucket.hensoldt-cyber.systems/projects/EXT/repos/s2n/browse> to TRENTOS-M and make is available via the TLS API
 - implement/port WireGuard for TRENTOS-M
 - Webserver native: lighttp or apache
 - ftp, MQTT server.  [SEOS-1924 - Authenticate](#) to see issue details
 - A [Honeypot](#) server would be a cool thing to have.  [SEOS-1924 - Authenticate](#) to see issue details
 - Demo app that uses network stack and NIC as lib  [SEOS-779 - Authenticate](#) to see issue details
 - Make MUX channel upper channel API also match exactly the lower API  [SEOS-132 - Authenticate](#) to see issue details
 - Extend Linux Proxy to implement a timer tick  [SEOS-329 - Authenticate](#) to see issue details
 - Implement timer driver based on Linux Proxy Tick  [SEOS-393 - Authenticate](#) to see issue details
 - Extend Linux Proxy to implement a RTC  [SEOS-1427 - Authenticate](#) to see issue details
- MiG-V and TRENTOS-M
 - RISC-V Debug/Trace: http://iis-projects.ee.ethz.ch/index.php/Trace_Debugger_for_custom_RISC-V_Core
 - add a MiG-V platform to QEMU
 - Run well established benchmark suites to get real-world performance numbers:
 -  [SEOS-1576 - Authenticate](#) to see issue details
 -  [SEC-153 - Authenticate](#) to see issue details
 -  [SEOS-1186 - Authenticate](#) to see issue details
 - Linux in MiG-V: [MiG-V v1.1 Linux](#)
- Port TRENTOS-M to various 3rd part platforms
 - extent [Raspberry Pi](#) support
 - implement more drivers
 - support RasPi4, see also <https://github.com/nspin/seL4/tree/rpi4> [Unknown User \(lukgra01\)](#) is working currently on porting the RPi 4 during his IDP @TUM
 - Odroid C2 (see [TRENTOS on ODROID-C2](#))
 - It's a reference platform for seL4 that is also used in the UNSW AOS course (<https://www.cse.unsw.edu.au/~cs9242/20/>)

- [Unknown User \(micpet01\)](#) plan to use it for [Virtualization](#), see also  [SEOS-1506 - Authenticate](#) to see issue details
 - Ultra96 Board (see [Avnet Ultra96](#))
 - run a demo in the ARMv8 cluster
 - optionally make a demo with something in the the FPGA
 - get the 4 port Ethernet breakout card running with the on-SOC MACs
 - PolarFire SoC (<https://www.microsemi.com/existing-parts/parts/152514>)
 - SiFive HiFive Unmatched (<https://www.sifive.com/boards/hifive-unmatched>)
 - rockpro64
- seL4/CAMkES related
 - explore CAMkES, write example demos using the available connectors, create a customized connector, improve / complete the existing public documentation
 - use LLVM to parse CAMkES files instead of doing this with a custom python script. Might be nicely aligned with using LLVM instead of gcc to build everything.
 - explore the MCS kernel
 - Create protocol design for cascaded channel muxers using (HDLC) nesting  [SEOS-104 - Authenticate](#) to see issue details
- Crypto related (identified by [Unknown User \(bendri01\)](#))
 - Add more crypto algorithms to crypto library
 - Add more TLS cipher suites
 - merge mbedTLS main line security fixes back to the version we are using
 - Test the entropy source on the MIG-V chip by reading out a lot of "randomness" and running it through a test suite (e.g., DIEHARD, DIEHARDER, NIST SP-800xx)
- Secure Update
 - with SUIT
 - <https://www.ietf.org/id/draft-ietf-suit-architecture-08.txt>
 - SUIT is already used in some projects (<https://github.com/wolfSSL/wolfBoot>, <https://github.com/ARMmbed/suit-manifest-generator>)
 - implement part, e.g. a manifest generator and parser
 - in Q1/2021 we have finished: [Kumar Gowtham - Secure Update](#),  [SEOS-827 - Authenticate](#) to see issue details
 - Research of the secure over the air update protocols (what's in the industrie etc.).
 - Doing backend of our secure update.
- CI, Test Automation and Repository Handling
 - CI Control board controlling another DevBoard
 - RasPi controller another RasPi
 - this is in progress now, see [CI Test Farm](#) where we use a MinnowBoard a test controller
 -  [CI-190 - Authenticate](#) to see issue details